

Documentation Tabs

Please open the provided documentation file on your laptop/PC. The file contains separate tabs for each data structure:

- Instruction
- Array
- Linked List
 - Singly Link List
 - Doubly Linked List
- Stack
- Queue
- Trees
- Sorting Algorithms

Each tab contains specific requirements and examples for the corresponding data structure implementation

General Requirements

Students are required to implement comprehensive programs covering the following core Data Structure topics:

- Arrays
- Linked Lists
- Stacks
- Queues
- Trees
- Sorting Algorithms

Each implementation must be menu-driven, allowing users to perform all standard operations relevant to each data structure.

Implementation Requirements

1. Full Implementation Required

Each program must include all standard operations pertinent to the respective data structure:

- Arrays: Insertion, deletion, display, etc.
- Stacks: Push, pop, peek operations
- Queues: Enqueue, dequeue, display functions
- Linked Lists: Insertion (beginning, middle, end), deletion, traversal
- Sorting: Multiple algorithms (Bubble, Insertion, Selection)

2. Menu-Driven Format

All programs must provide a clear, user-friendly menu interface. Example for Array operations:

===== ARRAY OPERATIONS MENU =====

1. Push to Stack
2. Pop from Stack
3. Peek Stack
4. Enqueue to Queue
5. Dequeue from Queue
6. Display Array
7. Delete Element (Pointer-Based)
8. Find Third Minimum Value
9. Eliminate Duplicates
10. Exit

3. File Structure

Submit one code file per topic, containing all related operations for that data structure. Use the following naming convention:

- <rollno>_Array.cpp
- <rollno>_Stack.cpp
- <rollno>_LinkedList.cpp
- <rollno>_Queue.cpp

Important Note: For each data structure, implement a single comprehensive program with different functions for each operation, integrated through the menu system.

4. Code Standards

- Include explanatory comments throughout your code
- Follow proper indentation and formatting practices
- Use meaningful variable and function names
- Implement appropriate error handling

Documentation Requirements

Dry Run Documentation

A manual dry run of each implementation is mandatory:

- Document step-by-step execution processes
- Record memory/stack/queue state changes
- Track variable values and transitions
- Include your full name and roll number on each page
- Submit clear images or scans compiled into a single PDF named `<rollno>_DryRun.pdf` or you can upload a dry run PDF for each data-structure.

Evaluation Process

Viva Examination:

Viva will be conducted to evaluate your understanding.

Be prepared to:

- Explain your implementation logic in detail
- Justify your dry run steps
- Demonstrate understanding of pointer usage and operations
- Address questions regarding performance considerations

Note: The assignment grade will be determined solely through the viva performance.